

Gentrification as Spatial Reallocation: Evidence on Urban Welfare and Distributional Effects

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Abstract

This paper examines gentrification as a process of spatial reallocation within urban systems, highlighting a central efficiency–equity trade-off: while gentrification is associated with improvements in aggregate economic conditions, its benefits are unevenly distributed and rising housing costs constrain welfare gains for low-income renters. Using a harmonized panel of U.S. census tracts from 1980–2010, we construct a composite gentrification index based on inflation-adjusted income, educational attainment, and housing costs, and estimate its relationship with neighborhood outcomes using pooled OLS with year fixed effects and a shift-share instrumental variables strategy.

The results indicate that gentrification is associated with higher median income, lower poverty and unemployment rates, increased rents, and rising skill concentration. Population levels increase in gentrifying tracts, consistent with spatial reallocation rather than aggregate displacement, although localized displacement of specific subgroups cannot be ruled out. Distributional analysis shows that income gains are attenuated in renter-dominated and economically vulnerable neighborhoods, while long-term opportunity measures—including child poverty and high school dropout rates—show no statistically significant response.

Instrumental variable estimates are positive but substantially smaller than OLS estimates, suggesting upward bias and reinforcing a cautious interpretation of causal effects. Overall, the findings support a view of gentrification as generating aggregate economic gains alongside meaningful distributional asymmetries.

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1 Introduction

Urban neighborhoods in the United States have undergone a profound spatial transformation over the past three decades. This period, spanning from 1980 to 2010, has been characterized by sharp increases in inflation-adjusted housing prices, the inflow of high-human-capital residents, and significant shifts in local amenities—a process collectively understood as gentrification. While these changes are often scrutinized at the micro-neighborhood level, they fundamentally represent a broader reconfiguration of the urban spatial organization. As households relocate in response to shifting economic signals, the city-wide distribution of population, local economic conditions, and exposure to opportunity are fundamentally reordered.

Most existing literature examines gentrification through a localized lens, focusing on immediate outcomes such as housing affordability, displacement, and crime. Although this approach has generated vital insights, it often abstracts from the interconnected nature of urban systems. In practice, neighborhoods are linked through migration flows, commuting patterns, and price adjustments, implying that a shock in one location propagates throughout the entire metropolitan area. Consequently, the welfare implications of gentrification cannot be fully evaluated without accounting for these broader general equilibrium interactions.

A growing body of research highlights the relevance of these mechanisms. Studies on gentrification document how amenity improvements—particularly safety and infrastructure—drive housing demand, while spatial equilibrium models emphasize how households sort across locations based on the trade-offs between wages, rents, and commuting costs. In parallel, the neighborhood effects literature demonstrates that a household’s location plays a critical role in shaping long-term economic mobility. Despite these advances, these strands remain largely disconnected, leaving limited evidence on how the long-run process of gentrification affects urban welfare once spatial reallocation and accessibility are internalized.

This gap is particularly critical in the modern U.S. context. Declining urban crime rates since the 1990s, the rising valuation of central-city living, and labor market restructuring have intensified spatial sorting. These forces suggest that gentrification may generate aggregate efficiency gains by optimizing the allocation of households relative to high-productivity job centers. However, rising housing costs and mobility constraints for low-income renters suggest a persistent efficiency–equity trade-off.

This paper reconceptualizes gentrification as a process of spatial reallocation within an urban system. Utilizing a harmonized longitudinal dataset of U.S. census tracts from 1980 to 2010, the analysis studies how amenity-driven neighborhood change induces residential sorting and alters access to employment opportunities. By tracking shifts in inflation-adjusted housing costs, educational attainment, and poverty dynamics, the paper provides empirical evidence on the mechanisms through which gentrification operates. Specifically, we evaluate its implications for both aggregate urban efficiency and the distributional welfare of diverse demographic groups.

Three important caveats frame the analysis from the outset. First, the gentrification index incorporates income, which is also an outcome variable in several regressions; this mechanical correlation warrants the

alternative index robustness check we implement (Section 11). Second, because consistent tract-level identifiers are not available for panel fixed-effects estimation, the empirical strategy relies on pooled OLS with year fixed effects and clustered standard errors, limiting the strength of causal identification. We employ a shift-share IV strategy to partially address endogeneity, but interpret all results as associations unless stated otherwise. Third, standard errors are clustered at the year level with only three clusters, which may yield unreliable inference; significance levels throughout should be interpreted with this in mind (see Section 13).

The paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 states the research questions. Section 4 outlines the contributions. Section 5 formalizes the hypotheses. Section 6 develops the theoretical framework. Section 7 describes the data and variable construction. Section 8 outlines the empirical strategy. Section 9 presents the main results. Section 11 reports robustness checks. Section 13 discusses limitations, and Section 14 concludes.

By linking longitudinal neighborhood change to city-wide reallocation, this paper contributes to a more integrated understanding of gentrification and its long-term consequences in modern urban economies.

2 Literature Review

Over the past three decades, U.S. cities have undergone substantial spatial transformation, with gentrification emerging as a central feature of urban change. While traditionally studied as a neighborhood-level phenomenon, a growing body of work suggests that gentrification reflects broader processes of spatial reallocation that reshape labor markets, accessibility, and welfare across cities. By utilizing a harmonized longitudinal dataset spanning 1980 to 2010, this paper evaluates how these spatial patterns persist over the long run and how they shift the urban equilibrium across three decades.

2.1 Neighborhood-Level Impacts of Gentrification

The foundational question in the gentrification literature is whether neighborhood upgrading displaces incumbent low-income residents. Early research emphasizes the risk that low-income households are displaced as higher-income residents move into previously disadvantaged neighborhoods (Freeman & Braconi, 2004). However, more recent empirical evidence using administrative and longitudinal data challenges the strength of this displacement narrative. Several studies find limited evidence of systematic displacement and show that many incumbent residents remain in gentrifying neighborhoods without experiencing significant adverse income effects (Brummet & Reed, 2019; Ding et al., 2016).

Consistent with this view, recent work using large-scale eviction records shows that most forced displacement occurs in persistently low-income neighborhoods rather than in those undergoing gentrification (Hepburn et al., 2024). Similarly, evidence tracking individual income trajectories finds little indication that gentrification negatively affects incumbent households' labor market outcomes or forces systematic relocation to worse neighborhoods (Beauregard, 2024).

A central mechanism emphasized in this literature is the role of endogenous amenity shifts. Improvements in neighborhood quality—often measured by proxy through rising educational attainment and declining poverty rates—are capitalized into housing prices and attract higher-income households. While foundational work has used crime reductions to proxy for these amenities (Ellen et al., 2017; Autor et al., 2017), recent scholarship suggests that the concentration of high-human-capital residents themselves acts as a primary amenity, driving further demand and neighborhood transformation. This feedback loop is central to our empirical framework.

2.2 Spatial Equilibrium and Urban Reallocation

Urban economics provides a complementary framework through spatial equilibrium models, in which households and firms sort across locations based on wages, rents, amenities, and commuting costs. In this framework, local shocks propagate through migration and price adjustments, affecting outcomes across the entire urban system (Roback, 1982; Moretti, 2011).

Empirical work highlights the importance of such reallocation mechanisms. Infrastructure expansions, particularly highways, have been shown to generate large population shifts and contribute to suburbanization (Baum-Snow, 2007). Similarly, transportation investments affect employment growth and the spatial distribution of economic activity across cities (Duranton & Turner, 2012). More broadly, spatial equilibrium models of local labor markets emphasize how localized shocks diffuse through migration, wages, and prices, ultimately shaping worker welfare across locations (Moretti, 2011).

Despite its strong theoretical foundation, this literature typically abstracts from neighborhood-level processes such as gentrification and does not directly evaluate how amenity-driven neighborhood change affects welfare within cities. Our framework bridges this gap by modeling gentrification as an amenity-driven shift in location-specific utility and deriving testable implications for housing prices, skill sorting, and welfare.

2.3 Neighborhood Effects and Economic Opportunity

A third strand of literature examines how neighborhoods shape long-term economic outcomes. Evidence from large-scale administrative data shows that children growing up in different neighborhoods experience substantial differences in income, education, and upward mobility, even within the same city (Chetty et al., 2018). These findings highlight the importance of access to high-opportunity neighborhoods in determining lifetime outcomes.

At the same time, research on labor market inequality documents increasing spatial sorting of workers across locations, particularly along skill dimensions (Autor, 2010). High-skill workers tend to concentrate in high-productivity areas, while low-skill workers face constraints such as high housing costs and limited mobility. These patterns suggest that spatial allocation plays a central role in shaping inequality and economic opportunity.

However, this literature typically treats residential mobility as exogenous and does not analyze how

processes such as gentrification endogenously drive relocation and exposure to different opportunity environments.

2.4 The Integration Gap

While these strands provide important insights, they remain largely disconnected. The gentrification literature focuses on localized housing and displacement outcomes; the spatial equilibrium literature emphasizes system-wide reallocation without incorporating neighborhood change; and the neighborhood effects literature highlights long-term consequences without modeling endogenous mobility.

Taken together, these studies suggest a common underlying mechanism: shifts in local human capital and economic opportunity attract higher-income households, inducing spatial reallocation that reshapes access to jobs across neighborhoods. Furthermore, while recent studies have utilized high-frequency administrative data to challenge the displacement narrative, they often focus on short-run transitions and modern snapshots.

This paper addresses this gap by integrating these strands into a unified framework. It reconceptualizes gentrification as a process of spatial reallocation and empirically evaluates how neighborhood change—proxied by longitudinal shifts in education, poverty, and inflation-adjusted housing costs—affects accessibility, mobility, and welfare across households within cities over a thirty-year horizon, from 1980 to 2010.

3 Research Questions

The overarching question guiding this paper is: *How does gentrification reshape urban welfare through spatial reallocation in U.S. cities over a thirty-year horizon, and how do these effects operate through mobility, neighborhood composition, and distributional channels across communities?*

This broad question is decomposed into five empirical sub-questions:

1. **Reallocation Mechanism.** Does gentrification—identified by changes in educational attainment and housing costs—induce endogenous re-sorting of households across neighborhoods, as reflected in shifts in skill composition and population levels?
2. **Mobility and Opportunity.** Is neighborhood upgrading associated with improvements in economic conditions and long-term opportunity, as measured by changes in poverty rates, child poverty, and educational outcomes?
3. **Distributional Effects.** Do the welfare effects of gentrification differ systematically across groups—particularly between higher-income and more vulnerable populations—proxied by assisted income receipt and housing tenure?
4. **Housing and Affordability.** Does gentrification generate rising housing costs and affordability pressures, as captured by changes in median rent and rent-to-income ratios?

5. **Aggregate Welfare.** Is gentrification associated with improvements in overall neighborhood economic performance, including income growth and reductions in unemployment?

4 Contributions

- **Reframing Gentrification as Spatial Reallocation:** This paper reconceptualizes gentrification as a process of spatial reallocation within urban systems rather than a purely localized phenomenon. By examining changes in neighborhood skill composition and population levels, we provide empirical evidence that gentrification is associated with endogenous sorting of households across neighborhoods.
- **Long-Run Empirical Evidence (1980–2010):** Using a harmonized tract-level dataset spanning three decades, we provide a long-run perspective on neighborhood transformation. This allows us to capture persistent changes in income, poverty, and housing markets across multiple economic cycles.
- **Evidence on Reallocation vs. Displacement:** By jointly analyzing changes in educational attainment and total population, we distinguish between displacement and reallocation mechanisms. The results indicate that gentrification is associated with increases in both skill concentration and population, suggesting reallocation rather than large-scale displacement at the aggregate level. However, we carefully note that this does not preclude localized displacement of specific subgroups.
- **Welfare Effects of Gentrification:** We provide systematic evidence on the relationship between gentrification and neighborhood welfare outcomes. The analysis shows that gentrification is associated with higher median income and lower unemployment and poverty rates, while having limited effects on long-term opportunity measures such as child poverty and educational dropout rates.
- **Housing Market and Affordability Impacts:** The paper documents strong housing market responses to gentrification, including significant increases in inflation-adjusted rents. By analyzing rent-to-income ratios, we highlight the potential affordability pressures that accompany neighborhood upgrading, with particular attention to the distributional asymmetry between renters and homeowners.
- **Distributional and Heterogeneous Effects:** Through interaction models, we show that the effects of gentrification are not uniform across neighborhoods. The results indicate heterogeneous impacts depending on levels of economic vulnerability and housing conditions, highlighting the unequal distribution of welfare gains.
- **Evidence on the Efficiency–Equity Trade-off:** The findings highlight a central trade-off in urban economics: while gentrification is associated with improvements in aggregate economic

outcomes, rising housing costs and heterogeneous impacts suggest unequal distribution of these gains across populations.

- **Integration of Three Literatures:** The paper integrates the gentrification, spatial equilibrium, and neighborhood effects literatures into a unified empirical framework, providing a more complete picture of how neighborhood change propagates through the urban system.

5 Hypotheses

While existing literature highlights localized effects of gentrification, this paper empirically evaluates its role in reshaping neighborhood composition, welfare outcomes, and housing markets over the long run. Utilizing the 1980–2010 harmonized panel, we formalize the research questions into six testable hypotheses:

- H1 (Amenity-Induced Housing Demand):** Gentrification, proxied by increases in educational attainment and income, is associated with higher housing demand, leading to statistically significant increases in inflation-adjusted median rent.
- H2 (Skill-Based Spatial Sorting):** Gentrification induces spatial re-sorting of households, leading to a significant increase in the share of residents with a bachelor’s degree in affected neighborhoods.
- H3 (Reallocation vs. Displacement):** Gentrification is associated with increases in total population at the aggregate tract level, indicating that spatial reallocation rather than net displacement is the dominant adjustment mechanism in the aggregate. This hypothesis does not preclude the possibility of localized displacement within subgroups that aggregate counts cannot detect.
- H4 (Welfare Effects):** Gentrification is associated with improvements in neighborhood economic outcomes, reflected in higher median income and lower poverty and unemployment rates.
- H5 (Distributional Effects and Housing Pressure):** Gentrification has heterogeneous effects across neighborhoods, with stronger impacts in areas with higher renter shares and economic vulnerability, and is associated with increased housing cost pressures as measured by rent and rent-to-income ratios.
- H6 (Limited Long-Run Opportunity Effects):** Gentrification has limited or statistically insignificant effects on long-term opportunity measures, such as child poverty rates and high school dropout rates, reflecting the sluggish transmission of neighborhood-level economic gains to intergenerational mobility. This result may partly reflect selection: families remaining in gentrifying tracts are not a random draw from the pre-gentrification population.

6 Theoretical Framework

Building on spatial equilibrium models of urban structure (Roback, 1982; Moretti, 2011) and evidence on infrastructure-induced spatial reallocation (Baum-Snow, 2007; Duranton & Turner, 2012), we model the city as a system of interconnected neighborhoods. Households choose locations based on a trade-off between labor market opportunities, housing costs, and local amenities.

The framework is a stylized spatial equilibrium model designed to guide empirical analysis across the 1980–2010 panel. In contrast to models where location characteristics are fixed, we model **gentrification as an endogenous amenity shock** that reconfigures housing demand and induces spatial reallocation. The model is designed to organize empirical analysis rather than to be taken to the data structurally.

6.1 Household Utility and Location Choice

Each household i chooses neighborhood $j \in J$ to maximize indirect utility:

$$U_{i,j} = w_j - r_j + A_j - c_{ij}, \quad (1)$$

where:

- w_j : real income opportunities, represented in our data by **inflation-adjusted median household income**.
- r_j : real housing costs, represented by **inflation-adjusted median gross rent**.
- A_j : local amenities, proxied by neighborhood **educational density** and the inverse of the **poverty rate**.
- c_{ij} : commuting costs between residence j and employment centers.

Households sort across neighborhoods such that, in equilibrium, no household can increase utility by relocating, conditional on housing prices and mobility constraints. Households with lower mobility costs—proxied here by higher education—are better able to exploit cross-neighborhood utility differentials.

6.2 Housing Market Equilibrium

In the short run, housing supply H_j is inelastic. Market clearing requires that the sum of households choosing location j equals the available supply:

$$\sum_i \mathbf{1}\{j(i) = j\} = H_j. \quad (2)$$

As higher-income households increase demand for neighborhood j , housing prices r_j adjust upward. This capitalization effect is a primary driver of the spatial reallocation of lower-income, rent-sensitive households.

6.3 Gentrification as an Endogenous Amenity Shock

We model gentrification as an exogenous increase in neighborhood-level amenities:

$$A'_j = A_j + \Delta A_j, \quad \Delta A_j > 0. \quad (3)$$

Empirically, this shift is identified by decadal increases in the share of residents with a **bachelor's degree** and concurrent **reductions in poverty**. This shock raises local utility, triggering three simultaneous effects:

1. Increased residential demand leading to rent capitalization ($\partial r_j / \partial A_j > 0$).
2. Endogenous sorting, as higher-income households move toward the treated neighborhood.
3. Spatial reallocation, as price-sensitive residents seek alternative neighborhoods with lower rents or higher accessibility. Critically, net population counts at the tract level may rise even as some incumbent low-income residents are displaced, if the volume of higher-income in-movers exceeds the volume of out-movers.

6.4 Welfare and Distributional Decomposition

Total welfare change (dW) resulting from neighborhood transformation can be decomposed into three primary channels:

$$dW = \underbrace{\sum_i dA_{j(i)}}_{\text{Amenity effect}} - \underbrace{\sum_i dr_{j(i)}}_{\text{Housing cost effect}} + \underbrace{\sum_i dw_{j(i)}}_{\text{Accessibility/income effect}}. \quad (4)$$

This decomposition highlights the **efficiency–equity trade-off**: while the amenity effect and accessibility-driven wage gains may increase aggregate welfare, the housing cost effect represents a transfer from renters to property owners, often resulting in negative welfare shocks for vulnerable, rent-burdened populations. The net sign of dW is thus heterogeneous across the income distribution and across tenure groups, motivating the distributional analysis in Section 9.

Empirically, the amenity effect is proxied by changes in educational density; the housing cost effect by changes in inflation-adjusted median rent (Table 8); and the income effect by changes in median household income (Table 2).

6.5 Testable Implications

The model yields three testable implications:

- **Amenity Capitalization:** Increases in neighborhood education levels lead to higher real housing costs (tests H1).

- **Spatial Sorting:** High-income households relocate toward neighborhoods with high A_j (tests H2 and H3).
- **Welfare Divergence:** Aggregate efficiency may increase while distributional outcomes vary significantly based on household tenure status and skill level, owing to the asymmetric housing cost effect in Equation (4) (tests H4–H6).

7 Data Sources and Variable Construction

7.1 Primary Data Source

The primary data source for this study is the **Geomarker Harmonized Historical Census Dataset (1980–2010)** (Geomarker, 2023). This longitudinal panel provides spatially consistent census tract-level data across four decennial census years, enabling the analysis of long-run neighborhood transformation and spatial reallocation. Spatial harmonization ensures that tract boundaries are held approximately constant across rounds, allowing longitudinal comparison of neighborhood characteristics. All monetary variables are adjusted to constant 2010 dollars using the provided inflation adjustment factors.

7.1.1 Neighborhood Demographics and Human Capital

- **Variables:** Total population (`population_total`) and population under age 18 (`population_under_18`).
- **Human Capital Markers:** Percentage of the population with no high school diploma (`pct_no_hs`), high school diploma (`pct_hs`), bachelor’s degree (`pct_bach`), and education levels higher than a bachelor’s degree (`pct_higher_than_bach`). High school dropout rates (`pct_hs_dropout`) are also included.
- **Use:** These variables are used to identify gentrification dynamics through changes in educational attainment and to capture shifts in human capital composition across neighborhoods.

7.1.2 Housing Market and Affordability

- **Variables:** Median household income (`median_income`, `median_income_2010_adj`), median gross rent (`median_rent`, `median_rent_2010_adj`), and median home value (`median_home_value`, `median_home_value_2010_adj`).
- **Housing Characteristics:** Vacancy rates (`pct_vacant_housing`) and homeownership rates (`pct_owner_occupied`).
- **Use:** These measures capture changes in housing affordability, cost pressures, and capitalization of neighborhood improvements into property values and rents.

Economic Opportunity and Welfare Proxies

- **Variables:** Percentage of population below the poverty level (`pct_poverty`), percentage of children under 18 below the poverty level (`pct_poverty_under_18`), and percentage of households receiving assisted income (`pct_assisted_income`).
- **Labor Market Indicators:** Unemployment rate (`pct_unemployed`).
- **Household Structure:** Female-headed households (`pct_female_hh`) and single-parent households (`pct_single_parent`).
- **Use:** These variables proxy for neighborhood economic conditions, welfare dependence, and social vulnerability, which are key to understanding distributional impacts of gentrification.

7.1.3 Urban Structure and Density

- **Variables:** Total land area in square meters (`area_m2`) and total population (`population_total`), used to compute population density.
- **Derived Measure:** Population density (population per square meter).
- **Use:** These measures control for differences in urban form, density, and spatial constraints that influence neighborhood change and housing market dynamics.

7.2 Key Variables

7.2.1 Gentrification Index

The gentrification index is constructed as a standardized composite of neighborhood socioeconomic status using three core dimensions: income, education, and housing costs. Specifically, we standardize each component and aggregate them into a single index:

$$\text{Gentrification}_{it} = Z(\text{Income}_{it}) + Z(\text{Education}_{it}) + Z(\text{Rent}_{it}), \quad (5)$$

where:

- $Z(\text{Income}_{it})$: Standardized inflation-adjusted median household income (`median_income_2010_adj`)
- $Z(\text{Education}_{it})$: Standardized share of population with a bachelor's degree (`pct_bach`)
- $Z(\text{Rent}_{it})$: Standardized inflation-adjusted median rent (`median_rent_2010_adj`)

This index captures cross-sectional variation in neighborhood socioeconomic status and serves as the primary explanatory variable across all empirical specifications. Consistent with the empirical results, higher values of the index are associated with significant increases in median income, declines in poverty and unemployment rates, strong increases in educational attainment (skill inflow), and rising housing costs, particularly rents.

Important caveat on mechanical correlation. Because income is included in both the gentrification index and in the primary outcome regressions, OLS estimates on median income may be upward biased through a mechanical channel. We address this directly in two ways: (i) an alternative gentrification index that excludes the income component (Section 11), and (ii) the IV strategy in Section 8.6, which isolates exogenous variation in gentrification. Results on income should be interpreted with this caveat in mind; results on poverty, unemployment, and other outcomes are less affected by this concern.

7.2.2 Spatial Reallocation and Skill-Based Sorting

Due to the absence of individual-level migration data, spatial reallocation is inferred through changes in neighborhood composition:

- **Skill Composition:** Measured using the percentage of residents with a bachelor’s degree (`pct_bach`). Regression results show a strong positive relationship between gentrification and skill concentration, indicating inflow of high-human-capital households.
- **Population Adjustment:** Total population (`population_total`) is used to distinguish between displacement and reallocation. Empirical results indicate a positive association with gentrification, suggesting population inflows rather than outflows. However, net inflow does not rule out displacement of low-income incumbents if they are replaced by a larger number of higher-income in-movers; this distinction motivates the distributional analysis below.
- **Socioeconomic Sorting:** Changes in assisted income (`pct_assisted_income`) and poverty rates (`pct_poverty`) serve as proxies for shifts in neighborhood composition along the socioeconomic dimension.

This approach aligns with spatial equilibrium models, where household sorting occurs along skill and income dimensions rather than direct demographic transitions.

7.2.3 Housing Pressure and Affordability

To capture the cost-of-living effects of gentrification, we construct the following measures:

$$\text{RentIncomeRatio}_{it} = \frac{\text{MedianRent}_{it}}{\text{MedianIncome}_{it}}, \quad (6)$$

$$\text{RenterShare}_{it} = 1 - \text{PctOwnerOccupied}_{it}. \quad (7)$$

Empirical results show that while income rises, rent increases substantially, generating heterogeneous effects across renter-dominated neighborhoods.

7.2.4 Welfare and Labor Market Outcomes

We evaluate welfare impacts using multiple outcome variables derived from the dataset:

- **Economic Outcomes:** Inflation-adjusted median income (`median_income_2010_adj`) and unemployment rate (`pct_unemployed`).
- **Poverty Measures:** Overall poverty rate (`pct_poverty`) and child poverty rate (`pct_poverty_under_18`).
- **Opportunity Proxies:** High school dropout rate (`pct_hs_dropout`).

Regression estimates indicate that gentrification is associated with increased income levels, reduced poverty and unemployment, and limited or statistically insignificant effects on long-term opportunity measures such as child poverty.

7.2.5 Urban Structure and Density Controls

To account for spatial characteristics, we construct population density as:

$$\text{PopDensity}_{it} = \frac{\text{Population}_{it}}{\text{Area}_i}, \quad (8)$$

using `population_total` and `area_m2`. Additional housing supply controls include vacancy rates (`pct_vacant_housing`) and owner-occupancy rates (`pct_owner_occupied`). We also construct a baseline eligibility dummy variable identifying initially low-income tracts based on baseline (1980) median income, used for robustness checks.

These controls ensure that estimated effects of gentrification are not confounded by differences in density, housing supply, or initial neighborhood conditions.

7.2.6 Instrumental Variables and Identification

To address endogeneity, we implement a shift-share instrumental variable strategy based on initial tract characteristics and metropolitan trends:

$$IV_{it}^{\text{skill}} = \text{InitialBach}_{i,1980} \times \text{MetroBachTrend}_t, \quad (9)$$

$$IV_{it}^{\text{density}} = \text{InitialDensity}_{i,1980} \times \text{MetroRentTrend}_t. \quad (10)$$

These instruments exploit exogenous variation in city-wide trends interacting with predetermined neighborhood characteristics. The exclusion restriction requires that initial tract characteristics affect outcomes only through their influence on subsequent gentrification, conditional on controls and year fixed effects. While plausible, this may be violated if initial characteristics also predict long-run trends in local productivity or amenities through channels other than gentrification.

A further concern, following [Goldsmith-Pinkham et al. \(2020\)](#), is that if the first-stage “shares” (initial tract skill composition and density) are correlated with pre-existing trends in neighborhood income trajectories, the exclusion restriction fails regardless of the validity of the metropolitan-level “shifts.” We cannot directly test this condition with the available data. While OLS results indicate strong positive effects of gentrification on income and welfare, IV estimates are smaller and statistically significant, but the magnitude is substantially smaller than OLS, suggesting potential endogeneity and highlighting the importance of cautious causal interpretation.

7.3 Control Variables

To isolate the effect of gentrification on neighborhood outcomes, we include a set of time-varying control variables capturing urban structure, economic conditions, and housing characteristics:

- **Urban Structure:** Population density (`pop_density`), capturing congestion effects and urban spatial structure.
- **Labor Market Conditions:** Unemployment rate (`pct_unemployed`), controlling for local labor market performance.
- **Socioeconomic Status:** Percentage of households receiving assisted income (`pct_assisted_income`), proxying for neighborhood economic vulnerability.
- **Housing Market Characteristics:** Percentage of owner-occupied housing units (`pct_owner_occupied`), capturing tenure structure and exposure to housing price changes.
- **Time Fixed Effects:** Year fixed effects (δ_t), implemented using `i.year`, control for aggregate macroeconomic shocks and time-specific trends affecting all neighborhoods.

In extended specifications, we also include interaction terms to capture heterogeneous effects:

- **Renter Exposure:** $\text{Gentrification}_{it} \times (1 - \text{PctOwnerOccupied}_{it})$
- **Housing Supply Constraints:** $\text{Gentrification}_{it} \times \text{PctVacantHousing}_{it}$
- **Vulnerability Interaction:** $\text{Gentrification}_{it} \times \text{PctAssistedIncome}_{it}$

These controls ensure that the estimated effect of gentrification is not confounded by underlying differences in labor markets, housing structure, or macroeconomic conditions.

8 Empirical Strategy

8.1 Identification Strategy

The empirical analysis exploits variation across census tracts and time using a pooled panel dataset spanning 1980–2010. Due to the absence of consistent tract-level identifiers suitable for fixed effects

estimation, we employ a **pooled OLS framework** with **year fixed effects**, which absorb aggregate macroeconomic shocks common to all tracts in a given census year (e.g., national house price cycles, labor market fluctuations, and inflation).

The identification strategy relies on cross-sectional and temporal variation in the constructed gentrification index, conditional on observable neighborhood characteristics. **However, with only three clusters (years 1990, 2000, and 2010 relative to the 1980 base), cluster-robust standard errors are known to perform poorly, and reported significance levels may overstate precision.** All significance claims should be interpreted with this caveat in mind.

8.2 Baseline Specification

The baseline model is:

$$Y_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}, \quad (11)$$

where Y_{it} represents outcome variables including median income, poverty rate, and unemployment rate; $\text{Gentrification}_{it}$ is the standardized index combining income, education, and rent; X_{it} includes control variables (population density, unemployment rate, percentage receiving assisted income, and percentage of owner-occupied housing); and δ_t represents year fixed effects. The coefficient β_1 captures the conditional association between neighborhood socioeconomic status and outcomes, net of observable controls and aggregate time trends. Standard errors are clustered at the **year level** to account for time-based correlation.

8.3 Spatial Reallocation and Skill-Based Sorting

To test H2 and H3, we estimate Equation (11) with Y_{it} set to pct_bach_{it} and $\text{population_total}_{it}$, respectively. A positive $\hat{\beta}_1$ on the bachelor's share indicates skill-based inflow; a positive $\hat{\beta}_1$ on total population indicates net in-migration at the aggregate level. As noted in H3, this aggregate result does not preclude localized displacement of specific subgroups:

$$\text{pct_bach}_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}, \quad (12)$$

$$\text{Population}_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (13)$$

8.4 Housing Affordability and Distributional Effects

To examine heterogeneous effects across tenure groups (H5), we augment the baseline with an interaction term:

$$\text{Income}_{it} = \beta_1 \text{Gentrification}_{it} + \beta_2 (\text{Gentrification}_{it} \times \text{RenterShare}_{it}) + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (14)$$

A negative $\hat{\beta}_2$ implies that income gains from gentrification are smaller in neighborhoods with a higher renter share, consistent with H5. We further examine housing pressure through:

$$\text{Rent}_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}, \quad (15)$$

$$\text{RentIncomeRatio}_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (16)$$

8.5 Opportunity and Welfare Outcomes

To assess broader welfare impacts:

$$\text{pct_poverty}_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}, \quad (17)$$

$$\text{pct_unemployed}_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}, \quad (18)$$

$$\text{pct_poverty_under18}_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (19)$$

8.6 Instrumental Variables (2SLS)

To address the endogeneity of the gentrification index, we implement a shift-share IV strategy. The first and second stages are:

$$\text{Gentrification}_{it} = \pi_1 IV_{it}^{\text{skill}} + \pi_2 IV_{it}^{\text{density}} + \gamma X_{it} + \delta_t + u_{it}, \quad (20)$$

$$Y_{it} = \beta_1 \widehat{\text{Gentrification}}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}, \quad (21)$$

where $\text{InitialBach}_{i,1980}$ and $\text{InitialDensity}_{i,1980}$ are baseline tract characteristics and MetroBachTrend_t and MetroRentTrend_t capture city-wide trends. IV estimates should be interpreted as informative but imperfect bounds rather than fully credible causal estimates.

8.7 Dynamic Specification

To capture persistence in neighborhood change, we include lagged values of the gentrification index:

$$Y_{it} = \beta_0 \text{Gentrification}_{it} + \beta_1 \text{Gentrification}_{i,t-1} + \beta_2 \text{Gentrification}_{i,t-2} + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (22)$$

Due to data limitations and the absence of tract-level panel structuring, lagged variables are constructed using sequential observations. Therefore, results from this specification should be interpreted as indicative of short-run persistence rather than strict causal dynamics.

9 Results

9.1 Descriptive Statistics

Table 1 reports summary statistics for the 168,699 tract-year observations spanning 1980–2010. The average tract has a population of approximately 4,274, an inflation-adjusted median income of \$45,434, and an inflation-adjusted median rent of \$782. The mean poverty rate is 14.3%, the bachelor’s degree share is 13.2%, and the unemployment rate is 4.7%. Substantial cross-tract dispersion across all variables reflects the pronounced heterogeneity of U.S. neighborhoods.

9.2 Baseline Welfare Effects (H4)

Table 2 presents OLS estimates of the baseline specification with median income as the outcome. The coefficient on the gentrification index is positive and highly significant: a one-standard-deviation increase in the index is associated with approximately \$8,253 higher inflation-adjusted median income ($p < 0.01$). The owner-occupancy share also enters positively, consistent with higher homeownership rates in more affluent and stable neighborhoods.

These estimates should not be interpreted causally. As discussed in Section 7, income appears both in the index and as the outcome, generating mechanical upward bias. The alternative index in Section 11 addresses this concern and yields qualitatively similar results. IV estimates (Table 12) further indicate that OLS overstates the magnitude of the effect by more than half (\$3,339 versus \$8,253).

Tables 3 and 4 show that gentrification is associated with statistically significant reductions in poverty ($\hat{\beta} = -1.79$, $p = 0.024$) and unemployment ($\hat{\beta} = -0.26$, $p = 0.042$), consistent with improved neighborhood economic conditions.¹ Time fixed effects reveal strong cyclicity, with poverty and unemployment rising sharply in 2010, likely reflecting the aftermath of the 2008 financial crisis.

9.3 Spatial Reallocation and Skill Sorting (H2, H3)

Table 5 shows that the gentrification index is strongly positively associated with the bachelor’s degree share ($\hat{\beta} = 3.19$, $p = 0.001$), indicating substantial inflows of high-human-capital residents. The high R^2 (0.76) suggests that educational composition is a primary margin along which neighborhood sorting occurs.

Table 11 reports a positive and statistically significant association between gentrification and total population ($\hat{\beta} = 154.3$, $p = 0.029$). The increase in population alongside rising skill shares is consistent with spatial reallocation rather than aggregate depopulation. However, this result should be interpreted cautiously. Net inflows do not preclude displacement of low-income renters if they are replaced by higher-income entrants. The estimate therefore supports the absence of *aggregate* displacement, not the absence of subgroup-level displacement. The low explanatory power ($R^2 = 0.030$) further suggests

¹Standard errors are clustered at the year level with only three clusters; p-values should therefore be interpreted as indicative rather than precise. The qualitative patterns are robust across specifications.

limited precision in explaining population dynamics. These findings are consistent with [Brummet & Reed \(2019\)](#) and [Hepburn et al. \(2024\)](#).

9.4 Housing Market Effects (H1, H5)

Table 8 shows that gentrification is associated with a statistically significant increase in inflation-adjusted median rent ($\hat{\beta} = 119.9$, $p = 0.002$), consistent with amenity capitalization. A one-standard-deviation increase in the index corresponds to approximately \$120 higher monthly rent (2010 dollars), or roughly 15% of the sample mean.

The rent-to-income ratio results in Table 9 are more nuanced. The negative coefficient suggests that income growth outpaces rent increases on average, implying no aggregate deterioration in affordability. However, this masks important heterogeneity. Table 6 shows that the interaction between gentrification and renter share is negative ($\hat{\beta} = -43.8$, $p = 0.099$), indicating attenuated income gains in renter-dominated neighborhoods. Given the limited number of clusters, this result should be interpreted as suggestive rather than definitive. Nonetheless, both the sign and magnitude align with the theoretical prediction in Equation (4): renters are disproportionately exposed to rising housing costs without benefiting from asset appreciation.

9.5 Opportunity and Long-Term Outcomes (H6)

Tables 7 and 10 examine child poverty and high school dropout rates as proxies for intergenerational opportunity. In both cases, coefficients are negative but statistically insignificant ($p = 0.158$ and $p = 0.342$), indicating limited evidence of improvement in long-term outcomes. These results are consistent with H6: short-run economic gains do not translate into measurable improvements in intergenerational mobility within the observed time horizon.

Several mechanisms may explain this pattern. Neighborhood composition evolves gradually, human capital investments respond with lags, and selection may play a role: households that remain in gentrifying areas are not a random subset of the original population.

9.6 Instrumental Variables (H4, Robustness)

Table 12 presents 2SLS estimates for median income. The IV coefficient remains positive but is substantially smaller than the OLS estimate (\$3,339 versus \$8,253), indicating upward bias in OLS arising from both mechanical correlation and reverse causality. As discussed in Section 8.6, the exclusion restriction may be violated if initial tract characteristics predict income growth through channels unrelated to gentrification (Goldsmith-Pinkham concern). Combined with the limited number of clusters, these results should be interpreted as informative but not definitive.

9.7 Summary of Findings

Overall, the results align closely with the welfare decomposition in Equation (4). The amenity effect—proxied by rising educational attainment—is strongly positive. The housing cost effect is substantial, with rents increasing by approximately \$120 per month. The income effect is positive at the aggregate level but attenuated in renter-dominated neighborhoods. Long-term opportunity measures show no significant response. Taken together, these findings support the presence of an efficiency–equity trade-off: gentrification is associated with aggregate economic improvements alongside uneven distributional outcomes.

10 Tables

Table 1: Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
year	168699	1996.076	12.411	1980	2010
population	168699	4274.435	2164.007	7	71894
population_u18	168699	1090.218	684.790	0	15299
median_inc	168699	37915.4	24698.08	2499	250001
median_inc_adj	168699	45433.62	27398.38	2499	254827
pct_poverty	168699	14.32855	12.3928	0	100
pct_poverty_u18	168699	12.83572	15.86691	0	100
pct_assistance	168699	5.285473	6.681969	0	100
pct_no_hs	168699	31.67619	25.00737	0	100
pct_hs	168699	43.65638	16.17792	0	100
pct_bach	168699	13.18132	9.453425	0	100
pct_higher_edu	168699	8.072662	7.956639	0	100
pct_hs_drop	168699	7.39644	10.91239	0	100
pct_female_hh	168699	18.42503	11.95226	0	100
pct_single_hh	168699	13.28412	9.57518	0	100
pct_unemployed	168699	4.699477	2.946336	0	48
pct_vacant	168699	9.449546	9.016681	0	95
pct_owner	168699	63.96938	22.58728	0	100
median_rent	168699	584.2927	389.6496	0	2001
median_rent_adj	168699	781.6021	350.6299	0	2001
median_home	168699	145126	151742.3	0	1000001
median_home_adj	168699	190887.5	156260	0	1000001
area_m2	168699	1.19e+08	1.29e+09	4975.345	2.23e+11

Notes: Summary statistics cover 168,699 tract-year observations from the Geomarker Harmonized Historical Census Dataset across census years 1980, 1990, 2000, and 2010. Monetary variables are in nominal dollars; regressions use 2010-adjusted values. Percentage variables range from 0 to 100. Substantial cross-tract dispersion in income, poverty, and educational attainment reflects the heterogeneity of U.S. neighborhoods.

Table 2: Effect of Gentrification on Median Income (Baseline OLS)

	Dependent variable: Median Income (2010 Adjusted)
Gentrification	8252.99*** (409.66)
Population Density	44783.86 (74424.91)
Pct Unemployed	-123.74 (124.10)
Pct Assisted Income	133.90 (154.74)
Pct Owner Occupied	237.18** (52.18)
Year FE	Yes
Constant	20681.32*** (2188.71)
Observations	168,699
R-squared	0.8955

Notes: OLS estimates of Equation (11). The dependent variable is inflation-adjusted median household income in 2010 dollars. The gentrification index is the standardized composite defined in Equation (5). Standard errors in parentheses are clustered at the year level (3 clusters); with so few clusters, reported significance levels may overstate precision. **Caution:** Because income is a component of the gentrification index, this coefficient likely overstates the true causal effect; see Table 16 for the alternative-index robustness check and Table 12 for IV estimates (\$3,339) that correct for this bias. *** p<0.01, ** p<0.05, * p<0.10.

Table 3: Poverty Model Regression Results

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	-1.7863	0.2793	-6.40	0.024	-2.9880	-0.5846
pop_density	188.8938	99.7857	1.89	0.199	-240.4492	618.2368
pct_unemployed	0.6819	0.1973	3.46	0.074	-0.1669	1.5307
pct_assisted_income	0.9061	0.1607	5.64	0.030	0.2146	1.5975
<i>Year Effects</i>						
1990	2.7719	0.8887	3.12	0.089	-1.0519	6.5956
2010	9.5659	1.0022	9.55	0.011	5.2539	13.8780
Constant	0.9494	1.2583	0.75	0.529	-4.4646	6.3634
Number of observations = 168,699						
R-squared = 0.6253						
Root MSE = 7.5858						
Robust standard errors clustered by year						

Notes: OLS estimates of Equation (11) with poverty rate as the dependent variable. The large positive year-2010 effect reflects aggregate poverty increases following the 2008 financial crisis, and should not be interpreted as evidence that gentrification worsens poverty; the gentrification coefficient remains negative and significant after controlling for this crisis-driven aggregate shock. Robust standard errors clustered at the year level (3 clusters). *** p<0.01, ** p<0.05, * p<0.10.

Table 4: Unemployment Model Regression Results

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	-0.2606	0.0552	-4.72	0.042	-0.4980	-0.0232
pop_density	30.0416	18.5483	1.62	0.247	-49.7652	109.8483
pct_assisted_income	0.2020	0.0177	11.39	0.008	0.1257	0.2783
<i>Year Effects</i>						
1990	0.0506	0.0701	0.72	0.546	-0.2511	0.3522
2010	2.4892	0.2133	11.67	0.007	1.5714	3.4070
Constant	2.5057	0.2122	11.81	0.007	1.5926	3.4187
Number of observations = 168,699						
R-squared = 0.3446						
Root MSE = 2.3853						
Robust standard errors clustered by year						

Notes: OLS estimates of Equation (11) with the unemployment rate as the dependent variable. The year-2010 fixed effect (2.49 percentage points) captures the aggregate labor market deterioration following the 2008 financial crisis and is not attributable to gentrification dynamics. Owner-occupancy share is excluded from this specification for parsimony; results are unchanged if included. Standard errors clustered at the year level. *** p<0.01, ** p<0.05, * p<0.10.

Table 5: Reallocation (Skill Change) Model Regression Results

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	3.1916	0.0776	41.15	0.001	2.8579	3.5253
pop_density	-10.1580	9.2286	-1.10	0.386	-49.8653	29.5494
<i>Year Effects</i>						
1990	-3.0062	0.1772	-16.97	0.003	-3.7685	-2.2440
2010	-1.4095	0.2486	-5.67	0.030	-2.4790	-0.3400
Constant	14.8437	0.1736	85.49	0.000	14.0966	15.5908
Number of observations = 168,699						
R-squared = 0.7615						
Root MSE = 4.617						
Robust standard errors clustered by year						

Notes: OLS estimates with the share of residents holding a bachelor's degree as the dependent variable. Because bachelor's share also enters the gentrification index, partial mechanical correlation is present; nevertheless, the strong positive association is consistent with endogenous skill sorting as predicted by the theoretical model. Year fixed effects are incorporated to control for time-specific shocks. Standard errors clustered at the year level. *** p<0.01, ** p<0.05, * p<0.10.

Table 6: Distributional Effects Model Regression Results

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	5439.531	804.402	6.76	0.021	1978.468	8900.593
interaction	-43.7518	14.9383	-2.93	0.099	-108.0261	20.5226
pop_density	-447880.2	49348.66	-9.08	0.012	-660210.3	-235550.0
pct_unemployed	-534.4902	230.7705	-2.32	0.147	-1527.415	458.4351
<i>Year Effects</i>						
1990	15536.3	737.4632	21.07	0.002	12363.25	18709.34
2010	10702.3	901.1692	11.88	0.007	6824.879	14579.72
Constant	38335.55	1471.182	26.06	0.001	32005.57	44665.54
Number of observations = 168,699						
R-squared = 0.8796						
Root MSE = 9505.6						
Robust standard errors clustered by year						

Notes: OLS estimates of Equation (14). The dependent variable is median household income (2010-adjusted). The interaction term captures heterogeneous effects of gentrification across neighborhoods with differing renter shares ($p = 0.099$, clustered over 3 year-groups), providing suggestive evidence that income gains are attenuated in renter-dominated neighborhoods, consistent with the distributional asymmetry in the theoretical decomposition (Equation (4)). Given the three-cluster inference problem, this result should be interpreted as directionally informative rather than precisely estimated. Standard errors clustered at the year level.

Table 7: Child Poverty (Opportunity) Model Regression Results

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	-2.1212	0.9619	-2.21	0.158	-6.2598	2.0174
pop_density	458.3488	182.1756	2.52	0.128	-325.4896	1242.1870
pct_unemployed	1.5009	0.7650	1.96	0.189	-1.7907	4.7925
<i>Year Effects</i>						
1990	5.5180	2.1704	2.54	0.126	-3.8206	14.8566
2010	-7.7190	2.2924	-3.37	0.078	-17.5823	2.1443
Constant	6.1119	5.4928	1.11	0.382	-17.5219	29.7456
Number of observations = 168,699						
R-squared = 0.4412						
Root MSE = 11.861						

Notes: OLS estimates with child poverty rate as the outcome. The negative but insignificant coefficient on the gentrification index ($p = 0.158$) indicates that short-run economic improvements in gentrifying neighborhoods do not translate into measurable reductions in child poverty within a single decade, consistent with H6. This null result may reflect genuine sluggishness in intergenerational transmission, or a selection bias: families remaining in gentrifying tracts are not a random draw from the pre-gentrification population. Standard errors clustered at the year level.

Table 8: Regression Results: Inflation-Adjusted Median Rent

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	119.8879	5.0724	23.64	0.002	98.0629	141.7129
pop_density	6508.247	1782.022	3.65	0.067	-1159.176	14175.67
year 1990	-68.3555	12.6344	-5.41	0.033	-122.717	-13.9940
year 2010	-54.3095	17.3164	-3.14	0.088	-128.816	20.1968
Constant	814.2933	15.2116	53.53	0.000	748.8432	879.7434
Observations	168,699					
R-squared	0.7828					
Root MSE	163.41					

Notes: OLS estimates with inflation-adjusted median rent as the outcome. A one-standard-deviation increase in the gentrification index is associated with approximately \$120 higher monthly rent, relative to a sample mean of \$782—a 15% increase. These results strengthen the evidence for heterogeneous distributional effects, as rising rents may offset income gains for certain groups and contribute to economic displacement risks. Negative year fixed effects for 1990 and 2010 partly reflect declining real rents during recessions. Standard errors clustered at the year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: Regression Results: Rent-Income Ratio

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	−0.0004615	0.0001846	−2.50	0.130	−0.001256	0.0003329
pop_density	0.5753363	0.1551803	3.71	0.066	−0.0923504	1.243023
year 1990	−0.0165011	0.0003499	−47.16	0.000	−0.0180066	−0.0149956
year 2010	−0.013916	0.0005155	−26.99	0.001	−0.0161342	−0.0116979
Constant	0.0313479	0.0003325	94.29	0.000	0.0299174	0.0327784
Observations	168,699					
R-squared	0.4939					
Root MSE	0.00778					

Notes: OLS estimates with the rent-to-income ratio as the outcome. The negative but insignificant ($p = 0.130$) gentrification coefficient indicates that income grows at least as fast as rent on average in gentrifying tracts, so the ratio does not worsen at the aggregate level. This result does *not* imply the absence of affordability pressures for specific groups; the distributional evidence in Table 6 shows that renter-dominated neighborhoods benefit less from income gains, and since renters cannot capture housing asset appreciation, their effective affordability may worsen even when the tract-level ratio is stable. Standard errors clustered at the year level.

Table 10: Regression Results: High School Dropout Rate

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	−0.9429	0.7628	−1.24	0.342	−4.2252	2.3393
pop_density	154.0326	73.3738	2.10	0.171	−161.6694	469.7346
year 1990	−0.0609	1.7756	−0.03	0.976	−7.7006	7.5787
year 2010	−11.0006	2.4786	−4.44	0.047	−21.6652	−0.3359
Constant	11.7026	1.8164	6.44	0.023	3.8872	19.5179
Observations	168,699					
R-squared	0.3820					
Root MSE	8.5784					

Notes: OLS estimates with high school dropout rate as the outcome. The insignificant coefficient on gentrification ($p = 0.342$) is consistent with H6. The large significant year-2010 effect (−11.0) reflects the secular national trend of declining dropout rates over the period, driven by broad educational policy changes rather than gentrification per se. As with child poverty, a selection concern applies: families who remain in gentrifying tracts may differ systematically from those who leave. Standard errors clustered at the year level.

Table 11: Regression Results: Total Population

	Coef.	Std. Err.	t	P> t	95% Conf. Interval	
gentrification	154.3401	26.9794	5.72	0.029	38.2573	270.4229
pop_density	13142.43	7682.242	1.71	0.229	−19911.59	46196.45
year 1990	−421.1107	56.0680	−7.51	0.017	−662.3519	−179.8696
year 2010	−455.3748	80.8188	−5.63	0.030	−803.1101	−107.6395
Constant	4584.289	40.8998	112.09	0.000	4408.311	4760.266
Observations	168,699					
R-squared	0.0297					
Root MSE	2131.7					

Notes: OLS estimates with total tract population as the outcome. The positive and significant gentrification coefficient indicates net population inflows in higher-indexed tracts at the aggregate level. This result should be interpreted narrowly as evidence against large-scale *aggregate* depopulation, not as evidence against displacement of individual low-income residents: net inflows are compatible with low-income renters being displaced and replaced by a larger number of higher-income in-movers, which tract-level counts cannot distinguish. The very low R-squared (0.030) further indicates that neighborhood socioeconomic characteristics explain very little variation in population levels, warranting caution. Year fixed effects capture the secular national population trend. Standard errors clustered at the year level. *** p<0.01, ** p<0.05, * p<0.10.

Table 12: 2SLS Estimates: Effect of Gentrification on Median Income

	Coef.	Std. Err.	z	p-value	95% CI	
Gentrification (instrumented)	3,339.2	37.887	88.14	0.000	3,264.9	3,413.4
Population density	−161,186.6	4,675.4	−34.48	0.000	−170,350	−152,023
Pct. unemployed	−3.254	9.490	−0.34	0.732	−21.853	15.345
Pct. assisted income	−274.243	5.626	−48.75	0.000	−285.270	−263.216
Constant	27,066.7	78.003	347.00	0.000	26,913.8	27,219.6
Observations	39,042					

Notes: 2SLS estimates of Equations (20)–(21), instrumenting the gentrification index with the skill-based and density-based shift-share instruments defined in Equations (9)–(10). The IV coefficient (\$3,339) is positive and significant but substantially smaller than the OLS estimate (\$8,253 in Table 2), suggesting meaningful upward bias in OLS attributable to both mechanical overlap between income and the index, and to reverse causality. Following Goldsmith-Pinkham et al. (2020), instrument validity requires that initial tract-level shares are uncorrelated with pre-existing outcome trends, a condition that cannot be directly verified. Reduced sample size reflects the subset of tracts with non-missing 1980 baseline data. Year fixed effects included.

11 Robustness Checks and Extensions

To validate the stability of the empirical findings and ensure that results are not driven by model specification, we implement a series of robustness checks aligned with the estimation strategy.

11.1 Spillover Effects

To assess whether local neighborhood change reflects city-wide dynamics rather than purely tract-level processes, we augment the baseline specification with a spillover measure defined as the deviation of the tract’s gentrification level from the period mean:

$$\text{GentrificationSpill}_{it} = \overline{\text{Gentrification}_t} - \text{Gentrification}_{it}. \quad (23)$$

Table 13 shows that the coefficient on the spillover measure is positive and significant ($\hat{\beta} = 2,932$, $p = 0.024$), indicating that tracts in periods with higher aggregate gentrification exhibit higher income even after controlling for own-tract gentrification. This is consistent with positive spillovers from city-wide neighborhood upgrading and allows us to assess whether neighborhood-level gains are influenced by broader general equilibrium trends.

11.2 Subsample Analysis: Baseline Eligibility

To verify that results are not driven by initially affluent tracts, we restrict the sample to tracts with below-median baseline income in 1980 (“eligible” tracts):

$$Y_{it} = \beta_1 \text{Gentrification}_{it} + \gamma X_{it} + \delta_t + \varepsilon_{it}, \quad \text{for Eligible}_i = 1. \quad (24)$$

Table 15 shows that the gentrification coefficient remains positive and significant ($\hat{\beta} = 4,121$, $p < 0.01$), though smaller than in the full sample, consistent with lower absolute income in these tracts. The results remain qualitatively consistent, indicating that the observed effects are not driven solely by high-income neighborhoods.

11.3 Alternative Outcome Variables

To assess robustness across different dimensions of neighborhood change, we estimate the model using alternative dependent variables including inflation-adjusted median rent, rent-to-income ratio, high school dropout rate, and total population. These specifications confirm that gentrification is associated with rising housing costs and changes in neighborhood composition, consistent with the baseline results.

11.4 Instrumental Variable Approach

The 2SLS results in Table 12 confirm a positive effect of gentrification on income, though with substantially smaller magnitude than OLS (\$3,339 vs. \$8,253). This pattern is consistent with upward OLS bias arising from the mechanical income component in the index and from reverse causality.

11.5 Alternative Gentrification Index

Table 16 constructs an alternative index that excludes the income component, using only educational attainment and rent:

$$\text{Gentrification}_{it}^{\text{alt}} = Z(\text{Education}_{it}) + Z(\text{Rent}_{it}). \quad (25)$$

The coefficient on income remains large and significant ($\hat{\beta} = 10,018$, $p < 0.01$), confirming that the baseline result is not mechanically driven by the inclusion of income in the index.

11.6 Log Specifications

Tables 18 and 17 report log-linear specifications for median income and median rent, respectively. The coefficients (0.151 for income, 0.137 for rent) indicate that a one-standard-deviation increase in the gentrification index is associated with approximately 15% higher income and 14% higher rent, confirming that results are robust to functional form and not driven by scale effects or outliers.

11.7 Heterogeneity Analysis

To examine whether the impact of gentrification varies across different neighborhood characteristics, we estimate interaction models:

$$\text{pct_poverty}_{it} = \beta_1 G_{it} + \beta_2 (G_{it} \times \text{PctAssisted}_{it}) + \beta_3 (G_{it} \times \text{PctVacant}_{it}) + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (26)$$

Both interaction terms are negative and significant at the 10% level ($\hat{\beta}_2 = -0.259$, $p = 0.015$; $\hat{\beta}_3 = -0.038$, $p = 0.079$), as shown in Table 14. This indicates that gentrification is more strongly associated with poverty reduction in neighborhoods with higher initial vulnerability, consistent with greater scope for improvement from a lower baseline, though mean reversion and compositional change offer alternative explanations. The results indicate heterogeneous impacts, with stronger effects observed in more economically vulnerable neighborhoods.

12 Tables: Robustness Checks

Table 13: Spillover Effects: Gentrification and Income

	Coef.	Std. Err.	<i>t</i>	<i>p</i> -value	95% CI	
gentrification	11539.2	40.02247	288.32	0.000	11367.0	11711.4
gentrification_spill	2932.25	466.757	6.28	0.024	924.0	4940.5
pop_density	−444095.3	146393.4	−3.03	0.094	−1073975	185784.7
pct_unemployed	−350.3342	186.9694	−1.87	0.202	−1154.8	454.1
year 1990	7899.655	159.9445	49.39	0.000	7211.5	8587.8
year 2010	omitted (collinear with spillover measure)					
_cons	45274.72	1226.147	36.92	0.001	39999.0	50550.4
Number of obs = 168,699						
R-squared = 0.8706						
Root MSE = 9854.9						
Std. Err. adjusted for 3 clusters in year						

Notes: Augmented specification including the spillover measure defined in Equation (23). The positive spillover coefficient indicates that aggregate city-wide upgrading contributes to local income even after controlling for tract-level gentrification, consistent with general equilibrium effects. The year-2010 indicator is omitted due to collinearity with the spillover term in this specification. Standard errors clustered at the year level.

Table 14: Heterogeneous Effects of Gentrification on Poverty Rate

	Coef.	Std. Err.	<i>t</i>	<i>p</i> -value	95% CI	
gentrification	−1.251066	0.3818986	−3.28	0.082	−2.894	0.392
interaction_assist	−0.25865	0.0315194	−8.21	0.015	−0.394	−0.123
interaction_vacancy	−0.0381217	0.0114125	−3.34	0.079	−0.087	0.011
pop_density	313.778	79.051	3.97	0.058	−26.350	653.906
pct_unemployed	0.8906512	0.1826016	4.88	0.040	0.105	1.676
Year: 1990	6.435202	0.9135762	7.04	0.020	2.504	10.366
Year: 2010	10.13449	1.413934	7.17	0.019	4.051	16.218
Constant	0.9857623	1.183265	0.83	0.492	−4.105	6.077
Observations = 168,699						
R-squared = 0.6346						
Root MSE = 7.4916						

Notes: OLS estimates of Equation (26). Both interaction terms are negative and significant at the 10% level, indicating that gentrification is more strongly (negatively) associated with poverty in initially more vulnerable and higher-vacancy neighborhoods. This could reflect mean reversion, greater scope for improvement from depressed baselines, or compositional change as higher-income households displace lower-income ones in vulnerable tracts. Standard errors clustered at the year level.

Table 15: Robustness: Subsample of Initially Low-Income (Eligible) Tracts

Variable	Coefficient	Std. Error	Significance
Gentrification index	4,121.05	(25.45)	***
Population density	−250,100	(5,120.71)	***
Unemployment rate	−136.42	(9.25)	***
Constant	28,127.71	(59.65)	***
Observations	39,042		
R-squared	0.686		

Notes: Subsample restricted to tracts with below-median baseline (1980) median income. The positive and statistically significant coefficient on gentrification remains robust within the subsample, indicating that income gains persist even when restricting the analysis to neighborhoods most likely to experience gentrification. The coefficient (\$4,121) is smaller than in the full sample (\$8,253), consistent with lower absolute income in these tracts. Year fixed effects included. *** $p < 0.01$.

Table 16: Robustness: Alternative Gentrification Index (Excluding Income Component)

Variable	Coefficient	Std. Error	Significance
Gentrification (alt. index)	10,017.90	(37.41)	***
Population density	−460,720	(9,871.13)	***
Unemployment rate	−749.26	(13.90)	***
Assisted income	−280.19	(6.75)	***
Observations	168,699		
R-squared	0.743		

Notes: The alternative gentrification index is constructed as $Z(\text{Education}) + Z(\text{Rent})$, excluding the income component. The large, significant positive coefficient on income confirms that the baseline result in Table 2 is not solely driven by mechanical correlation between the index and the outcome. This confirms that the baseline results are not driven by mechanical correlation between income and the gentrification measure. Year fixed effects included. *** $p < 0.01$.

Table 17: Robustness: Log Specification for Median Income

Variable	Coefficient	Std. Error	Significance
Gentrification index	0.1513	(0.0003)	***
Population density	−12.016	(0.183)	***
Unemployment rate	−0.0244	(0.0003)	***
Observations	168,699		
R-squared	0.870		

Notes: A one-standard-deviation increase in the gentrification index is associated with approximately 15% higher income. The log specification confirms that gentrification is associated with a significant proportional increase in income, indicating that the results are robust to functional form and not driven by scale effects or outliers. Year fixed effects included. *** $p < 0.01$.

Table 18: Robustness: Log Specification for Median Rent

Variable	Coefficient	Std. Error	Significance
Gentrification index	0.1371	(0.0003)	***
Population density	8.3182	(0.162)	***
Observations	167,253		
R-squared	0.728		

Notes: The log specification shows that gentrification is associated with a significant proportional increase in rent (approximately 14%), confirming that rising housing costs are robust to functional form and not driven by extreme values. The reduced sample reflects tracts with non-missing rent values. Year fixed effects included. *** $p < 0.01$.

13 Limitations

The findings of this paper should be interpreted in light of several important limitations.

Endogeneity and Causal Interpretation. The gentrification index is constructed from observed neighborhood characteristics that may be jointly determined with economic outcomes. While an instrumental variable strategy is employed, the estimates remain sensitive to potential violations of the exclusion restriction. The contrast between OLS and IV results suggests that causal interpretation should be approached with caution.

Shift-Share Instrument Validity. Following [Goldsmith-Pinkham et al. \(2020\)](#), the validity of shift-share instruments requires that initial tract-level shares are uncorrelated with pre-existing trends in the outcome variable. If tracts with high initial skill composition or density were already on steeper income trajectories before 1980—independent of gentrification—the instruments would fail the exclusion restriction even if the metropolitan-level trends are exogenous. This condition cannot be directly tested with the available data, and IV estimates should be understood as informative bounds rather than definitive causal estimates.

Mechanical Correlation in the Gentrification Index. Income is a component of the gentrification index and also appears as a dependent variable in several regressions. Although the alternative-index robustness check confirms the qualitative finding, all income-as-outcome results should be interpreted cautiously. Future work should consider constructing the index exclusively from non-outcome variables.

Limited Clustered Inference. Standard errors are clustered at the year level, resulting in a small number of clusters (three). This may lead to unreliable inference and overstated statistical significance, and therefore results should be interpreted cautiously. Wild cluster bootstrap or inference under few clusters would be more appropriate and is left for future work. Results that rest on borderline significance—notably the renter-share interaction ($p = 0.099$)—should be treated as suggestive rather than confirmed.

Pooled Cross-Section Rather Than Panel. The absence of consistent tract-level identifiers prevents the use of tract fixed effects, which would absorb time-invariant unobserved heterogeneity.

Unobserved sorting on time-invariant neighborhood characteristics—geography, historical land use, or institutional factors—may confound the estimated associations.

Dynamic Specification Limitations. Lagged variables are constructed using sequential observations rather than a fully structured panel framework. Consequently, the dynamic estimates should be interpreted as indicative of short-run persistence rather than precise intertemporal causal effects.

Selection Bias in Long-Run Opportunity Outcomes. The child poverty and dropout results reflect outcomes for families who remained in gentrifying tracts. These families are a selected sample—those with stronger place attachment, weaker outside options, or greater resources to withstand rising costs. The null findings on long-run opportunity may therefore reflect selection rather than a genuine absence of effects, and families who moved out may have experienced systematically different trajectories. Resolving this requires individual-level longitudinal data.

Incomplete Welfare Measurement. While the dataset includes key economic indicators such as income, poverty, and unemployment, it does not capture non-market amenities (park access, school quality, social capital), subjective well-being, commuting costs, or individual-level outcomes. These dimensions are central to the theoretical welfare decomposition in Equation (4) but cannot be operationalized with the available data.

14 Conclusion

This paper examines how gentrification reshapes urban welfare through spatial reallocation across neighborhoods in U.S. cities over the period 1980–2010. Using tract-level data, we construct a composite measure of gentrification and evaluate its relationship with economic outcomes, neighborhood composition, and housing markets, with a central focus on the efficiency–equity trade-off in urban development.

Three main findings emerge. First, gentrification is associated with improvements in aggregate neighborhood economic conditions. Gentrifying tracts exhibit higher median income, lower poverty rates, and lower unemployment rates, with results robust to alternative index construction and log specifications. However, IV estimates are substantially smaller than OLS—\$3,339 versus \$8,253—indicating that OLS estimates are upward biased by reverse causality, mechanical index construction, and potentially by pre-existing income trends not fully accounted for by the shift-share design.

Second, the dominant spatial adjustment mechanism appears to be reallocation rather than aggregate depopulation. Both the bachelor’s degree share and total population increase in gentrifying tracts, consistent with skill sorting and net in-migration. However, this aggregate pattern should be interpreted with caution: net inflows are fully consistent with localized displacement of low-income renters if they are replaced by a larger number of higher-income in-movers. The low explanatory power of the population regression ($R^2 = 0.030$) further tempers confidence in this result. These findings align with [Brummet & Reed \(2019\)](#); [Hepburn et al. \(2024\)](#) while preserving the possibility of subgroup-level displacement.

Third, distributional analysis indicates that aggregate gains are not uniformly shared. Gentrification is accompanied by rising rents, and income gains are attenuated in renter-dominated neighborhoods.

Although the rent-to-income ratio does not increase on average, renters bear the full burden of rising housing costs without benefiting from asset appreciation, implying a meaningful distributional asymmetry. Long-run opportunity measures, including child poverty and high school dropout rates, show no statistically significant response within the observation window, consistent with either slow intergenerational transmission or selection effects among households that remain.

Taken together, these findings characterize gentrification as a process that is associated with aggregate efficiency gains alongside unequal distributional consequences. Policies that promote neighborhood investment while protecting incumbent renters—such as rent stabilization, inclusionary zoning, and community land trusts—may help mitigate these distributional effects, though their implications for aggregate efficiency remain an open question.

Several extensions would strengthen the analysis. Access to individual-level administrative data would allow direct measurement of displacement and disentangle selection from true null effects in long-run outcomes. Incorporating measures of job accessibility and commuting costs would enable a more direct test of the mechanisms emphasized in the theoretical framework. Panel fixed-effects approaches using harmonized tract identifiers would improve causal identification, while alternative inference methods, such as wild cluster bootstrap, would address the limitations of few-cluster standard errors.

Overall, the evidence suggests that gentrification is best understood as a process of spatial reallocation with important implications for both aggregate welfare and the distribution of economic opportunity. These findings underscore the importance of evaluating urban transformation not only in terms of efficiency gains, but also in terms of how those gains are distributed across populations.

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